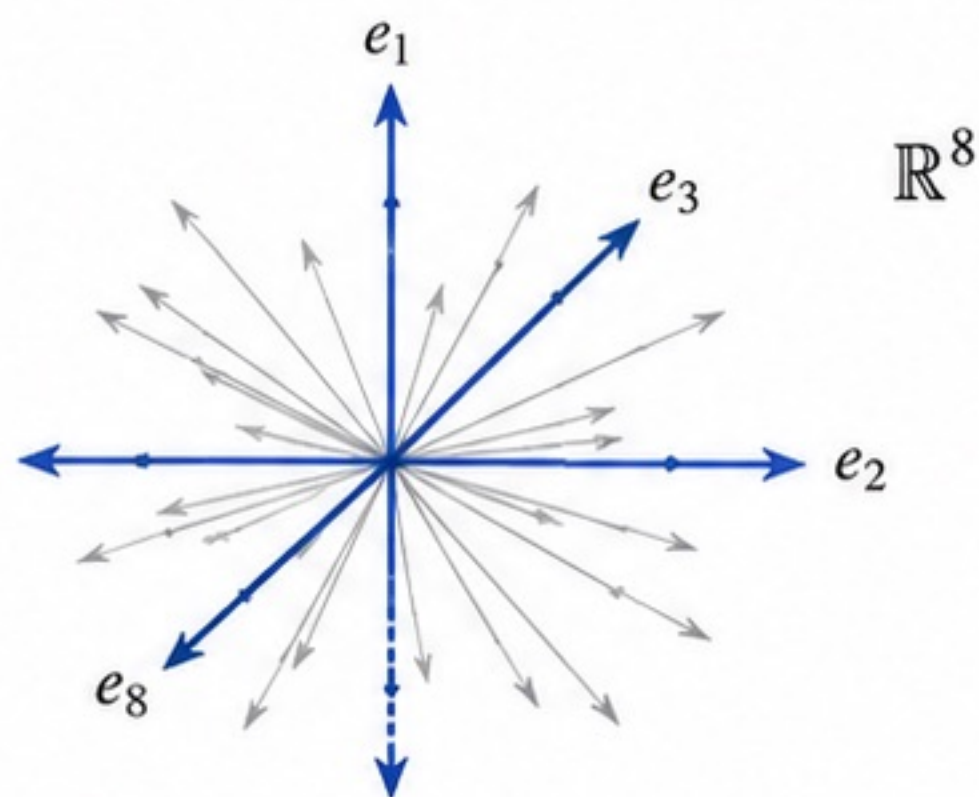


Observed residual floor is not an axis hole

Synthetic mixture: 80 features =
8 axis-aligned + 72 random directions

$$m = 8 \quad p = 0.05$$



— 8 axis-aligned directions
— 72 random directions

Protocol A: recovery vs N (trained TopK SAE, $d = 192$, $k = 8$)

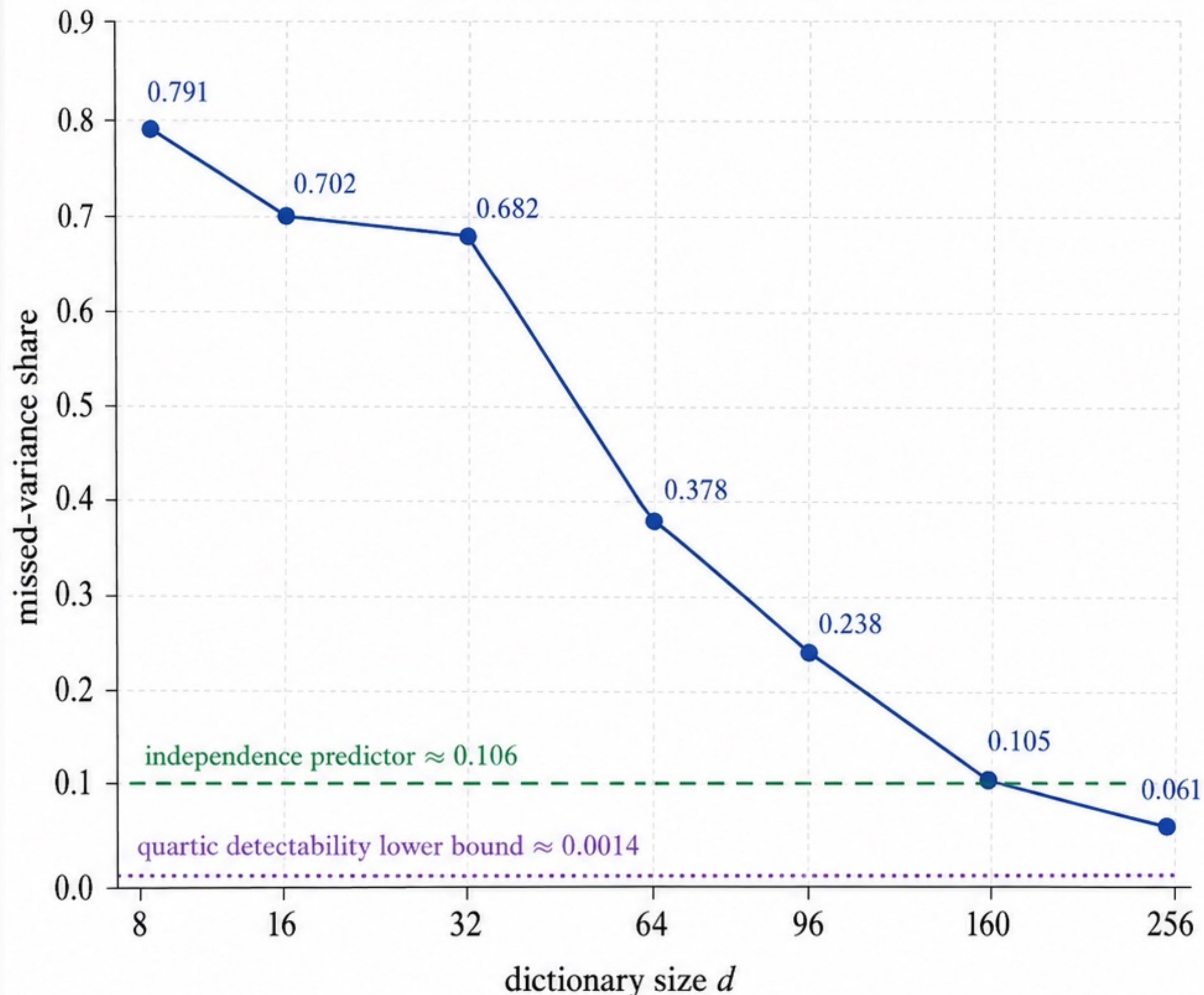
$N \in \{1024, 2048, 4096, 8192, 16384, 32768\}$

axis features	—●—	0.906
random directions	—●—	0.848

axis features account for $\approx 6\%$ of misses

*Protocol A message: axis features are not the
dominant source of misses.*

Protocol B: capacity sweep (trained TopK SAE, $N = 16384$, $k = 4$)



Capacity changes the observed residual floor; fixed-basis independence geometry does not explain the capacity trend.

Caution: Residual above the lower bound may reflect capacity, absorption, optimization, correlation, or nonlinearity rather than axis alignment.